Book Constructors and Destructor - C++ Program

```
#include <iostream>
#include <string>
#include <iomanip>
using namespace std;

class Book {
private:
    string title;
    string author;
    double price;

public:
    // Parameterized Constructor
    Book(string t, string a, double p) {
        title = t;
        author = a;
        price = p;
        cout << "[Constructor] Object Created: \"" << title << "\"\n";
    }

    // Copy Constructor
    Book(const Book &b) {
        title = b.title + " (Copied Edition)";
        author = b.author;
        price = b.price;
        cout << "[Copy Constructor] Object Copied: \"" << title << "\"\n";
    }

    // Destructor
    ~Book() {
        cout << "[Destructor] Destroying Object: \"" << title << "\"\n";
    }

    void display() const {
        cout << " Title : " << title << "\n";
        cout << " Author: " << author << "\n";
        cout << " Price : $" << fixed << setprecision(2) << price << "\n\n";
    }
};

int main() {
    cout << "=== Creating Original Book Object ===\n";
    Book book1("C++ Programming Essentials", "Bjarne Stroustrup", 59.99);
    cout << "\n--- Original Book Details ---\n";
    book1.display();
    cout << "=== Creating Copied Book Object (triggers Copy Constructor) ===\n";
    Book book2 = book1;
    cout << "\n--- Copied Book Details ---\n";
    book2.display();
    cout << "=== Program Ending (Destructors Auto-Invoked) ===\n";
    return 0;
}
```

Made with Julius